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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEZGYA01 – AN UNDERSTANDING OF RIVERINE ECOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A - ***Choose the Correct Option (Answer all questions) - Each question carries 1 marks***

1. A continuous record of discharge plotted against time is called

a). Stream flow b). Flow pulse

c). Hydrograph d). Flow regime **[U] / [CO1]**

2. In which course of river are potholes most typically found?

a). Lower Course b). Middle Course

c). Upper Course d). Delta Course **[U] / [CO1]**

3. The agricultural runoff leads to

a). Eutrophication b). Oxygen depletion in water

c). Decrease in aquatic organisms **d). All of the above** **[A] / [CO2]**

4. Stocking density means

a). Total fish weight **b). Number of fish per unit area**

c). Amount of feed d). Water depth **[U] / [CO3]**

5. The best time to visit Malarickal Village is during

a). June - September b). January - June

c). October - March d). April – July **[U] / [CO5]**

6. Which indicator best reflects biological State of a river?

a). Temperature trend **b). Species diversity**

c). Urban growth rate d). Industrial output **[U] / [CO2]**

7. Climate change affects rivers through _____.

a). Stable rainfall **b). Altered hydrology**

c). Reduced erosion d). Increased oxygen **[U] / [CO2]**

8. Which practice helps improve river water quality?

a). Dumping waste **b). Wastewater treatment**

c). Channel narrowing d). Over-extraction of water **[K] / [CO3]**

9. Which type of water body is most suitable for river cage culture?

a). Stagnant water with no flow b). Highly polluted rivers

c). Moderately flowing clean rivers d). Dry River beds **[U] / [CO3]**

10. Which of these is not a tributary of river Periyar?

a). Muthirapuzha b). Edamalayar

c). Kallar d). Mangalapuzha **[K] / [CO4]**

[1x10 = 10]

PART B ***Short Answer Questions (Answer any 14 questions - Each question carries 2 marks)***

11. Discuss how human activities causes hydrologic alteration in rivers. [U] / [CO1]

Human activities alter river hydrology by modifying flow regimes, sediment transport and channel morphology through dam construction and reservoir management, water withdrawal and diversion, urbanization, agriculture and deforestation, channelization and channel modification and climate change from use of fossil fuels. (Any 4 points)

12. What is environmental flow? [U] / [CO2]

Environmental flows (EFs) are the quantity, timing, duration, frequency and quality of flows required to sustain freshwater, estuarine, and near shore ecosystems and the human livelihoods and well-being that depend on them. **Liberal Valuation**

13. Mention any two regulating services of rivers. [K] / [CO3]

Water flow regulation, water purification and filtration, flood control, erosion control, ground water recharge, carbon sequestration, climate regulation etc.(Give marks if any two points written)

14. What is meant by River Bifurcation? [U] / [CO1]

River bifurcation is the splitting of a single river into two or more separate channels (distributaries) that flow downstream, often caused by topography, sediment build-up or tectonic activity. Bifurcation may be stable (Both channels receive water) or unstable (One channel eventually becomes dominant)

15. What are riverine ecosystem services? Give one example. [U] / [CO2]

Riverine ecosystem services are the essential benefits humans receive from healthy river systems, spanning provisioning (water, food), regulating (flood control, water purification), cultural services (recreation, spiritual value) and supporting services (nutrient cycling, sediment transport, habitat provision for aquatic and riparian species) **Definition with example**

16. Differentiate the catchment area and river basin. [U] / [CO1]

A catchment area is a specific, often smaller land area that drains water into a particular lake or reservoir or stream section. A river basin is the larger, comprehensive geographical region encompassing the entire area drained by a river and all its tributaries. It is composed of multiple smaller catchment areas or watersheds.

17. Define system state in the DPSIR framework in relation to river ecosystems. [K] / [CO2]

The 'state of the environment' is the combination of the physical, chemical and biological conditions prevailing in an environment as the result of pressure. Eg. Air quality (national, regional, local, urban, etc.) - Water quality (rivers, lakes, seas, coastal zones, ground water) - Soil quality (national, local, natural areas, agricultural areas) - Ecosystems (biodiversity, vegetation, soil organisms, water organisms) - Humans (health) - Soil use

18. What is meant by a regional or local river restoration model? [K] / [CO3]

The regional or local river restoration focus heavily on community-based, participatory approaches (often called **Nattukoottams** or people's gatherings) aimed at rejuvenating water bodies degraded by urbanization, sand mining, and pollution. These models often emphasize bio-restoration, sustainable livelihoods, and the revival of traditional water management practices.

19. What are allochthonous inputs? [U] / [CO1]

Allochthonous inputs refer to organic matter, nutrients or organisms imported from external surroundings. These inputs are vital food sources for fish and other organisms, especially in tropical streams. They also stabilize nutrient cycling and food webs.

20. What are the main pollution abatement works taken up by National River Conservation Plan? [U] / [CO4]

1. Interception and diversion works/ laying of sewerage systems to capture raw sewage flowing into the rivers through open drains and diverting them for treatment. 2. Setting up of Sewage Treatment Plants (STPs) for treating the diverted sewage 3. Construction of Low Cost Sanitation Toilets to prevent open defecation on river banks; 4. Construction of Electric Crematoria and Improved Wood Crematoria to conserve the use of wood; 5. River Front Development works, such as improvement of bathing ghats 6. Public participation & awareness and capacity building. (Any 2)

21. Give an account on aquatic larvae and adult insects in a riverine ecosystem. [U] / [CO1]

Aquatic insects in riverine ecosystem (mayflies, stoneflies and caddisflies) live as larvae in the water consuming algae and detritus and emerge as winged adults. Dragonflies and Damselflies – Their aquatic nymphs are predators of smaller aquatic animals and adults are aerial predators. Midges and mosquito larvae are crucial food sources of riverine ecosystem.

22. List any two effects of dams on streamflow. [U] / [CO2]

Dams alter streamflow by transforming natural variable river flow into regulated predictable patterns. Major effects are: Reduce peak flows (Flood control) increase low flows reversing natural seasonal flow patterns, trapping sediments, altering water temperatures, decreasing downstream flow. These effects vary in upstream and downstream sections of stream.

23. Why is ornamental fish farming popular? [U] / [CO3]

Growing Market Demand: The ornamental fish trade is experiencing significant growth, with a rising demand for diverse, colourful species. **High Profitability:** Ornamental fish farming is considered a lucrative, high-value industry, particularly because it can be established in small areas (backyard farming). **Technological Advancements:** Modern, easy-to-use aquarium equipment, such as automated filters and monitoring tools, has increased the popularity of keeping fish as pets. **Popularity as a Hobby:** Ornamental fish-keeping is the world's second most popular hobby. **Support for Small Entrepreneurs:** Many small farmers find this sector a viable business, with opportunities for growth through breeding high-demand species like guppies and platies**. (Any 2 points)**

24. What is Dissolved Oxygen (DO) and why is it important? [U] / [CO1]

Dissolved Oxygen is the amount of gaseous Oxygen (O_2) dissolved in water. It is essential for the respiration of aquatic organisms and overall ecosystem health. DO is important in aquatic life survival, water quality indicator, decomposition and nutrient cycling and in industries (Eg. For wastewater treatment)

25. Mention any two major constraints in cage culture. [An] / [CO3]

Major constraints in cage culture of fish include environmental degradation (poor water quality, low oxygen), rapid disease transmission due to overcrowding, high dependency on formulated feed, biofouling of nets, and logistical issues like poaching or site conflicts. These factors necessitate high-intensity management, regular monitoring, and increased operational costs.

26. What are invertivores? [U] / [CO1]

An invertivore is an animal that feeds primarily on invertebrates like insects, crustaceans, worms, molluscs and arachnids. They serve as a trophic link between aquatic invertebrates and larger species. Among the vertebrates fishes are the major invertivore animals in the stream ecosystem.

[2x14 = 28]

Part C ***Short Essay Questions (Answer any 8 questions - Each question carries 4 marks)***

27. Describe the Meenachil River MRRM model and its objectives. [U] / [CO2]

The Meenachil River & Rain Monitoring Network (MRRM) is a community-driven initiative in Kerala, using ~170+ rain gauges to monitor water levels and manage flood risks from Vagamon to Vembanad Lake. Key hydrological models applied to the basin include SWAT (Soil and Water Assessment Tool) for water quality and HEC-HMS for rainfall-runoff simulation.

Key Aspects of Meenachil River Modeling & Monitoring:

MRRM Network (Community Monitoring): Initiated in 2019, this network involves local volunteers (e.g., Meenachil River Protection Council) monitoring river gauges at locations like Erattupetta, Teekoy, and Kidangoor to provide real-time data.

Hydrological Models Used: SWAT (Soil and Water Assessment Tool): Applied to simulate streamflow, calibrate water quality, and analyze sediment/nutrient loads in the Meenachil River Basin (MRB).

HEC-HMS (Hydrologic Modeling System): Used to model flood scenarios in the sub-basin, achieving high accuracy (NSE > 0.7) in predicting peak flows.

Flood Management & Data: These models, along with the MRRM network data, are crucial for hazard mapping, flood forecasting, and operating local infrastructure like the Kanakkankadavu shutters to prevent inundation in agricultural areas. The river, spanning approximately 78 km, requires intense monitoring due to its flash flood susceptibility within the tropical humid zone of the Western Ghats.

28. Describe the Convention on Biological Diversity (CBD) and its objectives. [U] / [CO3]

The Convention on Biological Diversity (Biodiversity Convention or CBD) entered into force in December 1993, and has been ratified by almost 200 nations. The CBD is a 'Rio Convention' – it is one of three Conventions adopted at the "Earth Summit" (the Conference on Environment and Development), held in Rio de Janeiro, Brazil in 1992. The other two Conventions adopted were the UN Framework Convention on Climate Change), and the Convention to Combat Desertification.

As the first global treaty to provide a legal framework for biodiversity conservation, the Convention established three main goals: **1) the conservation of biological diversity 2) the sustainable use of its components3) fair and equitable sharing of the benefits arising from the use of genetic resources.**

Contracting Parties are required to create and enforce national strategies and action plans to conserve, protect and enhance biological diversity. They are also required to undertake action to implement the thematic work programmes on ecosystems and a range of cross-cutting issues which have been established to take forward the provisions of the Convention.

In January 2000 the ***Cartagena Protocol on Biosafety*** was adopted to supplement the provisions of the Convention. This Agreement, which came into force in September 2003, aims to ensure the safe handling, transport and use of living modified organisms (LMOs), resulting from modern Biotechnology.

Subsequently, the ***Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization to the Convention on Biological Diversity*** was adopted on 29 October 2010 and entered into force on 12 October 2014 (ratified by UK 22 May 2016). The Nagoya Protocol aims to share the benefits arising

from the utilization of genetic resources in a fair and equitable way, including by appropriate access to genetic resources and by appropriate transfer of relevant technologies, taking into account all rights over those resources, thereby contributing to the conservation of biological diversity and the sustainable use of its components. In October 2010, at the 10th Conference of the Parties to the CBD in Nagoya, Japan, the Parties adopted a new '*Strategic Plan for Biodiversity 2011–2020*' along with its 20 'Aichi targets'.

Parties to the CBD are required by Article 26 of the Convention to submit national reports to the Conference of the Parties on measures taken for the implementation of the Convention and their effectiveness in meeting the objectives of the Convention.

29. Classify mineral substrates in the riverbed based on partied size. [U] / [CO1]

Mineral substrates in riverbeds are classified based on particle size into categories ranging from fine mud to large boulders and bedrock. Their distribution is driven by flow velocity, transport capacity and geological sources. Generally riverbeds exhibit a downstream fining trend and coarser particles are found in upstream areas.

Based on common geological and ecological classifications riverbed substrates are divided into:

-**Bedrock (>512 mm):** Solid rock in the channel base – typical in high gradient steep streams

-Boulder (256-512 mm): Large heavy rocks, often found in mountain streams

-Cobble (64-256 mm): Smaller than boulders, provide stable and heterogeneous habitats

-Gravel (2-64 mm): Further divided into coarse gravel (16-64mm) and fine gravel (2-16mm). Generally gravel is the key substrate for fish spawning.

-**Sand (0.06-2mm):** Divided into coarse (0.5 – 2mm) and fine (0.06 -0.5 mm) sand. Often fills the gaps between larger stable substrates or forms mobile beds in lower gradient sections.

-Silt/Mud (0.06mm): Very fine particles that usually accumulate in low velocity areas. (The exact measurements used in Wentworth Scale may also be considered.)

30. Describe the major physical parameters used in river assessment. [U] / [CO2]

Major physical parameters to assess a river include channel morphology (width, depth, slope), discharge (flow volume), velocity, turbidity, water temperature, and total suspended/dissolved solids (TSS/TDS). These parameters define the river's energy, ability to transport sediment, and health, indicating habitat quality and identifying pollution sources.

Key Physical Assessment Parameters

Channel Morphology: Width and Depth: Key in determining the cross-sectional area and the hydraulic capacity. **Gradient (Slope):** Determines the energy, speed, and erosion type (vertical or lateral). **Channel Profile:** Longitudinal changes from source (steep) to mouth (gentle) and cross-sections. **Pools and Riffles:** Shallow areas (riffles) and deep pools indicate habitat diversity.

Hydraulic Features: Discharge (Q): Total volume of water passing a point over time, crucial for flood assessment. **Velocity:** Speed of flow, often measured at different depths and locations. **Thalweg:** The deepest part of the channel, marking the line of fastest flow.

Water Physical Properties: Temperature: Influences chemical and biological processes, influenced by climate and shading. **Turbidity:** A measure of water clarity, affected by suspended particles.**Total Suspended Solids (TSS) & Total Dissolved Solids (TDS):** Indicate sediment loading and ionic concentration, respectively. **Colour and Odour:** Basic sensory parameters indicating pollution or sedimentation issues.

Geomorphological Features: Meanders/Curvature: Bends in the river that indicate erosion on the outer bank and deposition on the inner bank. **Sediment Type:** Substrate composition (e.g., sand, gravel, rocks) in the riverbed.

Riparian Cover: Vegetation cover along the riverbanks, critical for shading and bank stability. These parameters are generally measured over time to detect changes in the river's ecological health.

31. Describe Strahler stream order method. [U] / [CO1]

Strahler stream ordering is a method for classifying streams based on hierarchical tributaries. The smallest, unbranched, permanent streams are designated as 1st order. When two streams of the same order (N) join, they form a higher order stream (N+1). Merging different orders, retains the higher orders.

Key Principles: 1st Order: Headwater streams with no tributaries. Increases by 1: Only when two streams of the same order meet. (1+1=2, 2+2=3)

Different Orders: If a 1st order stream joins a 2nd order stream, the channel remains 2nd order. This method is used to characterize watershed morphology, including drainage density, stream frequency and bifurcation ratio. It helps to identify headwaters (1 to 3 orders), intermediate streams (4 to 6 orders) and larger river (7+). Often used to define legal, environmental or regulating restrictions for building dams or managing water rights.

33. Explain how riverfront development measures can be applied to reduce pollution and improve water quality in an urban river. [A] / [CO3]

Riverfront development (RFD) measures can be applied to urban rivers to drastically reduce pollution and improve water quality by integrating nature-based solutions, upgrading infrastructure, and restoring natural ecosystems along the banks. Successful projects, such as the Sabarmati in Ahmedabad and various Ganga Riverfront developments, have shown that transforming neglected, polluted banks into managed green spaces can stop the direct disposal of waste and filter runoff.

Here are the key measures and how they improve water quality:

1. Ecological Restoration and Nature-Based Solutions

Riparian Buffer Zones: Establishing green buffers (often 75 meters wide) of native vegetation, trees, and wetlands along the banks acts as a natural filter, trapping sediments and absorbing nutrients from surface runoff before they enter the water. **Bio-swales and Rain Gardens:** These landscaping elements are designed to capture, treat, and infiltrate storm water runoff from urban surfaces, reducing the inflow of pollutants like oil, heavy metals, and suspended solids. **Constructed Wetlands:** These act as bio-filters, removing pollutants such as nitrogen and phosphorus from urban runoff through plant absorption and microbial activity. **Ecological Floating Islands:** Artificial floating islands with aquatic plants can be placed in slower-moving urban streams to absorb pollutants and increase oxygen-dependent microorganisms that improve self-purification.

2. Sewage and Wastewater Management

Interceptor Sewers: RFD projects often include the construction of interception systems that intercept raw sewage from existing, neglected drains and divert it to treatment plants (STPs) rather than allowing it to flow directly into the river. **Upgrading Treatment Infrastructure:** Setting up additional STPs along the riverbanks allows for the treatment of domestic and industrial wastewater, directly reducing the Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) in the river.**Closure of Open Drains:** Converting open, sewage-carrying drains into managed, covered pipelines that lead to treatment facilities eliminates illegal dumping and odour problems.

3. Structural and Bank Improvement

De-silting and De-concretization: Removing accumulated contaminated sediment from the riverbed and removing hard, unnecessary concrete structures allows the river to breathe and increases its self-purification capacity. **Soft**

Landscaping and Erosion Control: Replacing broken concrete retaining walls with gabion structures and native vegetation fixes the banks, prevents excessive sedimentation (erosion), and provides habitats for beneficial aquatic organisms. **Riverbank Stabilization:** Using natural materials like wood, stones, and vegetation helps stabilize banks and enhances the habitat for fish and beneficial microbes.

4. Non-Structural and Management Practices

Solid Waste Management Infrastructure: Installing organized masonry bins and regular cleaning services along the riverfront removes waste that would otherwise be dumped into the river by users. **Water Quality Monitoring:** Implementing regular water quality monitoring points within the developed riverfront allows for timely detection of pollution sources and assessment of the improvement measures. **Community Engagement and Education:** Actively involving local residents in cleanup efforts, educating them on the use of environmentally friendly products, and restricting polluting activities, such as direct disposal of waste during rituals, is crucial for long-term improvement.

Riverfront development is no longer just about beautification; it is a vital tool for ecological rejuvenation. By adopting a **"Nature-based Solution"** approach and prioritizing the ecological function of the river over purely aesthetic concrete structures, urban riverfront development can turn degraded water bodies into healthy, sustainable ecosystems.

34. Describe the Pen culture practices carried out in Kerala state. [U] / [CO5]

Pen culture in Kerala is primarily utilized as a climate-resilient technology for rearing fish fry to fingerlings in reservoirs, wetlands, and waterlogged areas, rather than just in the main flowing rivers themselves. This technique involves fencing shallow parts of the water body using bamboo poles and HDPE nets to allow for natural food availability while protecting the fish.

Key pen culture practices in Kerala include:

Seed Rearing (Nursery Management): Primarily used for raising fry to advanced fingerlings before releasing them into the larger water body (reservoir/wetland), which ensures higher survival rates.

Species Cultivated: Key species include Indian Major Carps *(Catla*, Rohu, Mrigal), Grass Carp, and Pearlspot (*Etroplus suratensis)* **Integrated Systems:** Pen culture is often integrated with clam farming in brackish water wetlands, where nets hold the clams in place during cultivation. **Site Management:** HDPE pens are installed in shallow regions of water bodies like the Vazhani and Malampuzha reservoirs. **Flood Mitigation:** It has been identified as a reliable method to withstand environmental crises, such as the 2018 flood, by protecting fish stock, as seen in studies of Black clam culture. These initiatives are often spearheaded by the Department of Fisheries and the Kerala State Co-operative Federation for Fisheries. Development (Matsyafed) as part of inland fish production enhancement, often in collaboration with research institutes like ICAR-CIFRI and CMFRI.

35. Explain the methods used for microbiological analysis of river water. [U] / [CO2]

Microbiological assessment of river water involves testing for fecal indicator bacteria—such as *E. coli*, total coliforms, and enterococci—to determine pollution levels and pathogen risks from sewage or agricultural runoff. Techniques include membrane filtration, multiple-tube fermentation (MPN), and agar plating to count colony-forming units (CFU), with levels often compared against WHO or local standards for drinking or recreational safety.

Key Microbiological Parameters and Indicators

Fecal Indicator Bacteria (FIB): *Escherichia coli* , fecal coliforms, and intestinal enterococci are used to indicate the presence of pathogens. **Pathogenic Microorganisms:** Detection of specific harmful organisms like *Salmonella*, *Vibrio cholerae*, and protozoa. **Total Viable Count (TVC):** Measures the overall bacterial population, indicating organic pollution levels.

Assessment Methodologies

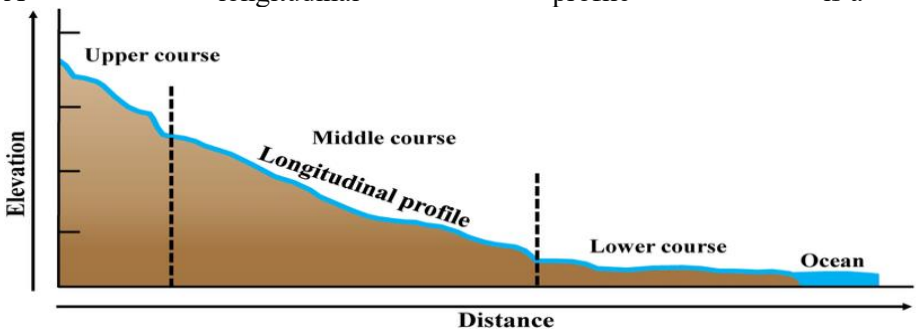
Sample Collection: Aseptically collecting water in sterile containers. **Serial Dilution:** Tenfold dilution (e.g., in Ringers solution) to enable colony counting. **Membrane Filtration Technique:** Passing water through a filter, then incubating on selective media. **Most Probable Number (MPN):** A statistical method to estimate the number of viable bacteria. **Incubation:** Culturing samples (commonly 22°C and 37°C) to isolate specific bacteria.

Factors Affecting Microbiological Quality

Pollution Sources: Urban wastewater, industrial waste, agricultural runoff, and livestock waste increase microbial loads. **Environmental Factors:** Rainfall, water temperature, flow rate, and season affect bacterial survival. **Self-purification:** Natural ability of the river to reduce contaminants **Impacts of contamination**: High concentrations of *E. coli* (>2000 MPN/100ml) indicate severe contamination, rendering water unsafe for recreational use. Contaminated water can cause illnesses like cholera, giardiasis, and dysentery, especially in developing regions. Routine monitoring is essential for identifying pathogens and protecting public health.

36. What is longitudinal profile of a river? Give its key characteristics. [U] / [CO1]

A longitudinal profile is a graph showing the elevation changes of a riverbed along its entire course, from the source (highest elevation) to the mouth (lowest elevation/base level). It typically forms a smooth concave curve, being steep in the upper course and flattening out towards the lower course, representing a balance between erosion and deposition.



Characteristics of a River Profile

Concave Shape: The profile is generally concave upward, representing a steady state of equilibrium where a river transports its sediment load efficiently.

Upper Course (Steep Slope): Located near the source in high-relief areas. Characterized by vertical erosion, steep gradients, fast flow, and features like rapids and waterfalls.

Middle Course (Moderate Slope): Gradient decreases, resulting in less vertical erosion and more lateral erosion/transportation.

Lower Course (Gentle Slope): Very gentle slope near the mouth. Characterized by deposition, resulting in floodplains and deltas.

Key Concepts in Longitudinal Profiles

Graded Profile: A smoothed, balanced profile where energy allows for the transport of sediment load.

Base Level: The lowest point a river can erode to, usually sea level, though lakes can be local, temporary base levels.

Knickpoint: A sharp change in the slope of the profile (e.g., a waterfall or steep rapid), often signaling a previous change in sea level or geological uplift.

Factors Influencing the Profile

Erosion and Deposition: The key processes that shape the profile, with steeper parts favoring erosion and flatter parts favoring deposition.

Base Level Changes: A drop in sea level can create more steepening, while a rise in sea level can lead to increased deposition.

Geological Structure: Hard and soft rocks can cause unevenness, known as an ungraded profile, which will erode into a smoother shape over time.

[4x8 = 32]